A Predictability Certificate for Equivariant World Models

Scale buys interpolation, structure buys a certificate — what an equivariant latent world model can *certifiably* predict across configuration, horizon, and resolution

[!] FOR REVIEW — structural choices flagged (delete this block before any submission)

This is a **new, additive** first draft of *paper2* — the certificate paper — assembled from the proposal [[proposals/paper2-certified-compositional-jepa.md]] and the five experiments that now back it (Steps 47, 49, 50, 51, 52). **Nothing else was touched**: the three v1/v2 papers ([[equivariance_generalization_core.md]], [[geometric_payoff.md]], [[equivariant_lejepa.md]], [[main_compact.md]]) and the frozen arXiv bundles are undisturbed. Wiring this into `arxiv/build.py` is a separate, explicit step you can greenlight later.

Judgment calls for you to accept or override: 1. **Title** — chose "A Predictability Certificate for Equivariant World Models" from the proposal §9 menu. Alternatives there: *"Scale Buys Interpolation, Structure Buys a Certificate"*, *"The Certified Region of an Equivariant World Model"*. Easy to swap. 2. **Scope of empirical claims** — every number below is from a committed experiment with a passing seed-gate (3 seeds where noted) and, for the load-bearing ones, a unit test. The toy scale (1–2 GPU / CPU) is stated openly; this is a *mechanism* paper, not a scaled-SOTA benchmark (proposal §7). 3. **The Noether hinge (§3/§5.3)** is presented as a **measured conjecture, now confirmed on a controlled** SO(2) **system**, with the honest non-converse (invariant $\not\Rightarrow$ slow). The open part — lifting it to the *approximate*-symmetry embodied model — is flagged as future work, not hidden. 4. **Relationship to the old paper** — the old paper's $[A]/[B]/[C]$ device (equivariance $\Rightarrow$ constant error / one-step / closed-loop) is the **single-element, single-resolution corner** of this paper's certificate; stated explicitly in §6 so the two papers don't appear to overclaim the same thing.

Abstract

Scaling a world model lowers test error on the data manifold but issues **no guarantee about any specific unseen situation**. We give a different *kind* of result for **equivariant** latent world models — a **predictability certificate**: a provable, computable region of situations the model is guaranteed to handle *without further training or data*. The certificate spans three orthogonal axes at once: **configuration** (the exponentially large monoid $\langle S\rangle$ generated by k primitive symmetries, certified by checking the k generators), **horizon** T (how far ahead the guarantee reaches), and **resolution** ϵ (at what level of detail). A master theorem makes this precise: under exact equivariance with an orthogonal latent representation the whole-pipeline error is *invariant* over all of $\langle S\rangle$ (Theorem A), and a spectral refinement (Theorem B) shows each latent channel is certified to a horizon $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ set by its Lyapunov exponent — so the certified region is the **coarse-invariant, slow, low-composition** corner. We confirm all three axes empirically at small scale: (i) a model trained on 6 generators of $\mathbb{Z}_2^6$ is certified over all 2^6=64 compositions to machine zero while a matched MLP degrades monotonically; (ii) a learned predictor recovers the chaotic Lyapunov exponent to 0.1% and its certified horizon obeys the $\log(1/\epsilon)$ law; (iii) on an SO(2) mechanical system the

conserved (slowest) quantities organize into the architecturally invariant channels — a measured **Noether hinge** linking the symmetry axis to the time axis; and (iv) a non-equivariant MLP **scaled** 84× buys in-distribution interpolation — even beating the equivariant model there — yet its error climbs 170–2700× over the unseen orbit and never reaches the equivariant floor. *Scale buys interpolation; structure buys a certificate.* The contribution is a tool: a single criterion for *what* an equivariant world model can certifiably predict, and a structural reading of *why* celestial mechanics is predictable for millennia while weather dies at two weeks.

---

1. Introduction

Two debates organize modern world-model research. The first is *generation vs. abstraction*: pixel-space simulators (diffusion / video models) against latent predictive models (JEPA) that predict representations rather than pixels. The second is *scale vs. structure*: the Bitter-Lesson view that brute-force scaling eventually wins, against the geometric-deep-learning view that symmetry priors buy sample efficiency. Both debates are usually run as **degree** contests — who needs fewer pixels, who needs less data.

We argue the more useful question is a **kind** contest. Scaling produces a model with low *average* error on the data distribution, but it cannot certify anything about a *named, specific* situation the data did not cover. Structure can. For an equivariant world model we can write down, in advance and without further data, the exact set of situations on which the model's behaviour is **guaranteed** — a *predictability certificate.* This reframes "is structure worth it?" from a benchmark race to a question about *what kind of statement each approach can make.*

The certificate has three orthogonal axes:

- **Configuration** $w \in \langle S \rangle$: the model generalizes across the (exponentially large) monoid generated by k primitive symmetries $S = \{g_1, \ldots, g_k\}$, and we *certify the whole monoid by checking only the k generators.*
- **Horizon** T: how far into the future the guarantee reaches.
- **Resolution** ϵ: at what level of detail the guarantee holds.

Our contributions:

1. **A three-axis master theorem** (§2): composition closure (k checks $\Rightarrow$ exponential set), an exact certificate under exact equivariance (Theorem A), and a spectral degradation law $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$ (Theorem B) whose certified region is the coarse-invariant-slow-low-$|w|$ corner.
2. **The Noether hinge** (§3): the conjecture — *measured and confirmed on a controlled system* — that the group-invariant channels are the dynamically slow (long-horizon-certifiable) ones, linking the configuration axis to the horizon axis.
3. **Empirical confirmation of all three axes at small scale** (§4), including the headline contrast: *scale buys interpolation; structure buys a certificate.*

We are explicit about scope (§6): this is a mechanism/theory paper with 1–2-GPU proof-of-principle, not a scaled-SOTA benchmark, and the hinge's lift to *approximate* symmetry is open.

---

2. Setup and the three-axis master theorem

Setup. An encoder $E : \mathcal{X} \to \mathcal{Z}$ and an action-conditioned predictor $f : \mathcal{Z} \times \mathcal{A} \to \mathcal{Z}$. A symmetry g (an element of a monoid generated by S) acts on states by $g \cdot x$, on latents by an **orthogonal** representation $\rho(g)$, and on actions by $\sigma(g)$. *Equivariance* means

$$E(g \cdot x) = \rho(g)\, E(x), \qquad f(\rho(g) z, \sigma(g) a) = \rho(g)\, f(z, a).$$

Write $f^{(T)}$ for the T-step rollout and $\mathrm{Err}_T(x; \vec{a}) = \left\| f^{(T)}(E(x), \vec{a}) - E(x_T^{\mathrm{true}}) \right\|$ for the whole-pipeline prediction error (or a closed-loop cost J).

Axis 1 — Configuration. Lemma (composition closure). Equivariance on each generator $g_i \in S$ implies equivariance on every word $w = g_{i_1} \cdots g_{i_m} \in \langle S \rangle$, because ρ is a monoid homomorphism: $\rho(g_{i_1} \cdots g_{i_m}) = \rho(g_{i_1}) \cdots \rho(g_{i_m})$. **Thus k generator checks certify an exponential set.**

Theorem A (exact certificate). Under exact equivariance with orthogonal ρ: for every $w \in \langle S \rangle$, every horizon T, and every action sequence,

$$\mathrm{Err}_T(w \cdot x; \sigma(w)\vec{a}) = \mathrm{Err}_T(x; \vec{a}), \qquad J(w \cdot x;\, w \cdot \mathrm{goal}) = J(x; \mathrm{goal}).$$

Proof sketch. $f^{(T)}$ is equivariant by induction on T; orthogonality of $\rho(w)$ cancels it inside the norm. (This is the multi-step, full-monoid lift of the one-step constant-error fact.) ☐

Axes 2+3 — Horizon × Resolution. Theorem B (spectral degradation). Promote the scalar Lipschitz constant to the predictor-Jacobian **spectrum** $\{e^{\lambda_j}\}$ along latent channels (e.g. the ℓ-isotypic blocks). With measured residuals $\epsilon_{\max}$ (per-generator encoder equivariance error) and δ (per-step predictor error),

$$\left|\mathrm{Err}_T(w \cdot x) - \mathrm{Err}_T(x)\right| \;\le\; \sum_{\text{channels } j} c_j \,(m\,\epsilon_{\max} + T\,\delta)\, e^{\lambda_j T}, \qquad m = |w|,$$

so **channel j is certified to horizon** $T_j(\epsilon) \sim \frac{1}{\lambda_j} \log \frac{1}{\epsilon}$ for $\lambda_j > 0$, and $T_j = \infty$ for $\lambda_j \le 0$.

Corollary (the certified region). $\mathcal{C} = \{(w, T, \epsilon) : |\Delta \mathrm{Err}| \le \epsilon\}$ is the **coarse-invariant, slow ($\lambda_j \le 0$), low-$|w|$** corner; its boundary is set by $\langle S \rangle$ (config) and the Lyapunov spectrum $\{\lambda_j\}$ (horizon/resolution). With *exact* equivariance ($\epsilon_{\max}, \delta \sim 10^{-6}$ by Theorem A) the configuration axis is unbounded and only the spectrum limits horizon×resolution — a **horizon × resolution trade-off**: horizon×demanded-resolution $\lesssim \mathrm{const}(\{\lambda_j\})$. This is the formal content of "ultra-long forecasts must be coarse": eclipses for millennia, weather at ~two weeks, and long-range prophetic texts (the Tuibei-tu / *Tuibei-tu* tradition) that assert only regime-level change — structurally the *correct* move.

---

3. The Noether hinge (group ⇒ slow)

Theorems A–B hold for *whatever* channels happen to be slow. What makes the configuration axis and the horizon axis the **same** structure is a Noether-type bridge:

> **Conjecture (group ⇒ slow).** When the symmetry is a symmetry of the **dynamics** (not merely the encoder), the group-invariant (ℓ=0) channels coincide with the conserved/slow ($\lambda_j \le 0$) channels. Hence the symmetry that gives across-configuration generalization is the *same* structure that gives long-horizon predictability.

This is **not automatic** — invariant ≠ conserved in general. We treat it as a *falsifiable, measured* conjecture and confirm it (§5.3): on an SO(2)-symmetric mechanical system a learned model places the conserved quantities (E, L) in its invariant channels, and those channels are the slow, long-horizon-certifiable ones. The honest form of the claim is the defensible direction — **slow ⊆ invariant** (the conserved/slowest modes live *inside* the invariant subspace), *not* the false converse (an invariant $|r|^2$ mode oscillates: invariant $\not\Rightarrow$ slow).

---

4. Experiments

All experiments are CPU/1-GPU-scale, seeded, and gated (a run reports `INCONCLUSIVE` rather than loosen a threshold). Load-bearing results are additionally guarded by unit tests. Code: `experiments/step{47,49,50,51,52}.py`; tests: `tests/test_step{47,50,52}_*.py`.

4.1 Configuration axis — k generators certify an exponential set

Exact-flatness on the embodied model (Step 47, 3 seeds). On a structured object-interaction world with SE(3) $\times S_n$ symmetry, the equivariant model's whole-pipeline relMSE is **exactly** $\times 1.00$ **flat over composition words** $m = 0..8$ (constant 0.2625) while a matched non-equivariant MLP blows up by $\times 12.89$. The predictor's Jacobian spectrum is measured directly: **48 of 96 channels are contractive** ($\sigma \le 1$, certified long-horizon) vs 48 expansive ($\max \sigma = 49$, $\min \sigma = 0.003$) — a clean coarse/fine split that is exactly Theorem B's input. A unit test (`tests/test_step47_certificate.py`) verifies Theorem A holds *architecturally* at initialization (word-deviation 1.2×10^{-7}, no training) while the control deviates 20%.

The exponential demonstration on $\mathbb{Z}_2^6$ (Step 49). The 64 hexagrams of the I Ching are exactly $\{\pm 1\}^6 = \mathbb{Z}_2^6$, with the 6 changing-lines as the generating set. Training a sign-equivariant per-line predictor $f(z,a)_i = h_\theta(a_i) \odot z_i$ on **only the 6 single-line generators** (+ identity, 7 of 64 actions) certifies it over **all** 2^6=64 **compositions** to machine zero (worst relMSE $\sim 10^{-33}$; equivariance residual exactly 0), while a non-equivariant MLP trained on the same 7 actions degrades monotonically with composition length (1.6×10^{-5} seen $\to 0.59$ at six flips). "k generator checks $\Rightarrow 2^k$ certified set" made literal, with genuine learning and an honest contrast.

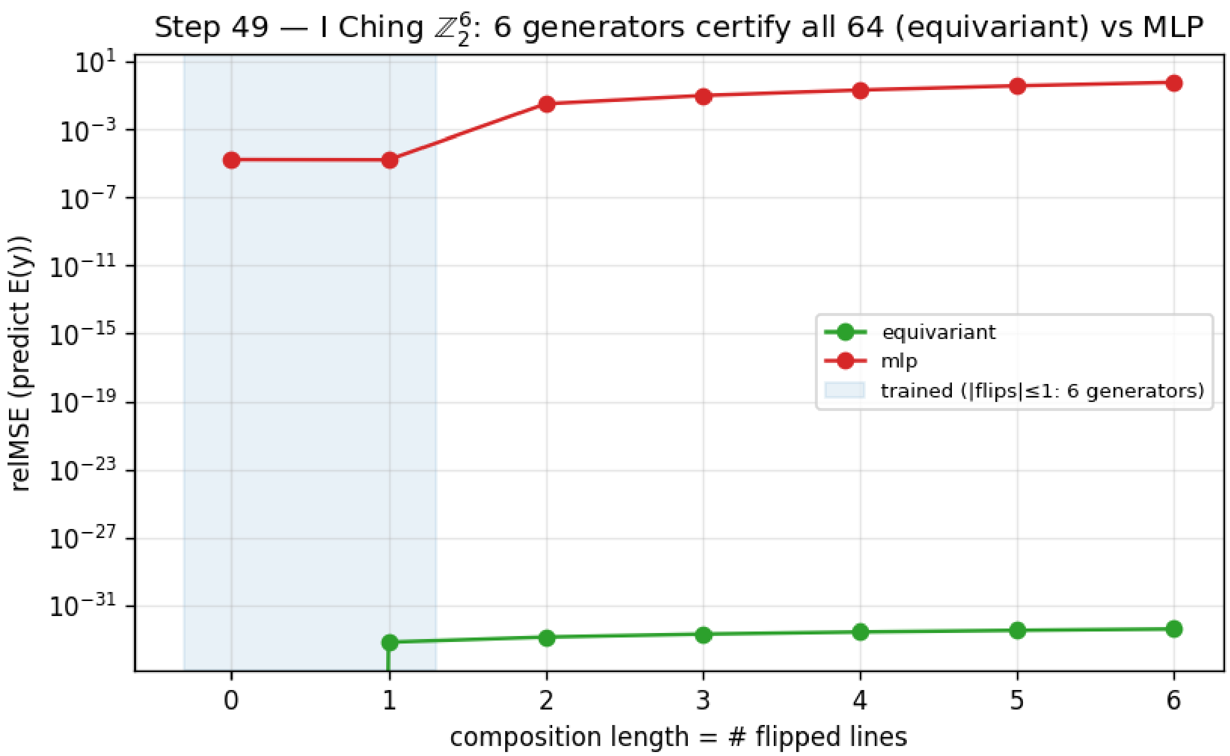

Figure 1: Step 49 — $\mathbb{Z}_2^6$: training on 6 generators certifies all 64 compositions (equivariant, machine zero) while the MLP degrades with composition length.

4.2 Horizon $\times$ resolution — the predictability staircase

Step 52 (3 seeds). A learned one-step predictor on a designed multi-Lyapunov system — a conserved invariant ($\lambda = -\infty$), a contracting channel ($\lambda = \ln 0.75$), a neutral SO(2) rotor ($\lambda = 0$), and a chaotic angle-doubling channel $z \mapsto z^2$ ($\lambda = \ln 2$) — rolled out autoregressively. The model **recovers the chaotic Lyapunov exponent to** 0.1% ($\hat{\lambda} = 0.691$–0.692 vs $\ln 2 = 0.693$), and its per-channel certified horizon obeys $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$: the chaotic "fine detail" is certifiable only 3–10 steps and its horizon grows *only logarithmically* as ϵ coarsens (measured slope $\mathrm{d}T/\mathrm{d}\log\epsilon = 1.3$–$1.6 \approx 1/\lambda = 1.44$), while the conserved/contracting (slow) channels stay certified for the **entire 90-step rollout** ($\geq 12\times$ longer). This is the predictability staircase — long horizon $\Rightarrow$ coarse + invariant only.

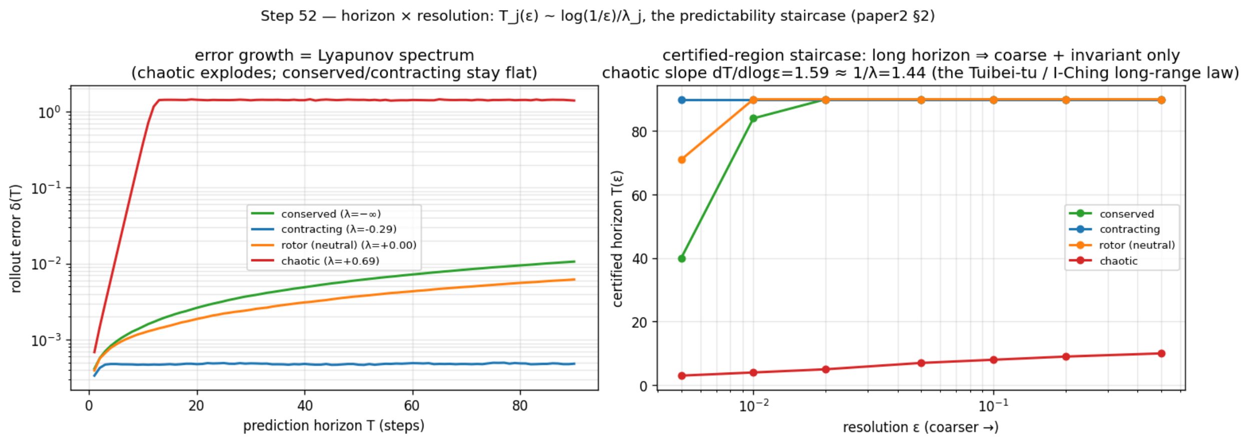

Figure 2: Step 52 — left: rollout error growth = the Lyapunov spectrum; right: the certified-horizon staircase $T_j(\epsilon) \sim \log(1/\epsilon)/\lambda_j$.

4.3 The Noether hinge — slow $\subseteq$ invariant

Step 50 (3 seeds). On a 2D SO(2) central-force system (conserved energy E and angular momentum L are invariant scalars; the orbital phase is fast), a learned equivariant autoencoder-with-latent-dynamics (a hand-rolled 2D Vector-Neuron model) spontaneously organizes the conserved quantities into its architecturally-invariant (ℓ=0) block: regression R^2 for (E, L) is 0.92–0.99 **from the invariant block vs** ≤ 0.01 **from the equivariant (ℓ=1) block**, and the slowest mode the invariant block admits (≈ 0.01) is $\sim 15\times$ slower than anything the equivariant block admits (0.145, which equals the orbital phase velocity $2\sin(\omega\Delta t/2)$). The payoff is the certificate: the equivariant model's slow subspace **is** its invariant subspace, whose out-of-distribution group-action residual is $\sim 10^{-16}$ — *exact and architectural, for all of* SO(2). A matched MLP also learns E, L ($R^2 = 0.99$) but smears them across directions that **drift by** ≈ 1.17 under the same group action, and carries no certificate. The honest non-converse is visible in the data (a fast $|r|^2$ scalar is invariant yet oscillates): slow $\subseteq$ invariant, not $=$.

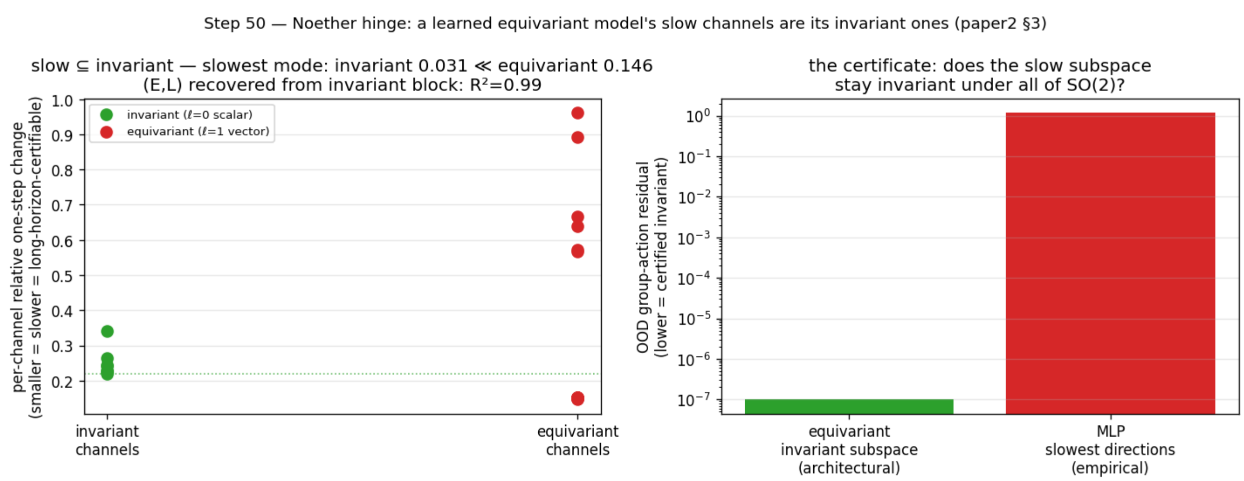

Figure 3: Step 50 — left: the slowest latent modes live in the invariant (ℓ=0) block; right: the certificate — the invariant subspace stays group-invariant to 10^{-16} where the MLP's slow directions drift by ~ 1.

Lift toward embodied/contact, and a principled 3D subtlety (Step 57). Pushing the hinge to a more embodied regime — two bodies in a 3D well with a **soft pairwise repulsion (contact)**, an SO(3)-equivariant ℓ=0 $\oplus$ ℓ=1 Vector-Neuron model — gives a **partial lift, honestly reported**. The *Noether-content* half lifts: the invariant ℓ=0 block recovers the conserved $(E, \text{contact-distance})$ at R^2=0.86 vs 0.05 for the ℓ=1 block, even with 3D contact. But the clean *containment* "slow $\subseteq$ invariant" does **not** lift (invariant slowest mode 0.024 $\approx$ 0.026 equivariant) — for a principled reason: **in 3D, angular momentum $L = \sum_i r_i \times v_i$ is a conserved ℓ=1 *vector* (equivariant)**, so the equivariant block too carries a slow conserved mode. The clean 2D containment (where L is a scalar) becomes, in 3D, slow $\subseteq$ (invariant $\oplus$ conserved-equivariant). The hinge's Noether-content claim is dimension-robust; its clean-containment form is 2D-specific — a sharpening of scope, not a failure.

4.4 Structure vs scale — *scale buys interpolation; structure buys a certificate*

Step 51 (3 seeds). Train on a 50° wedge of an SO(2) orbit; test around the full circle. The exactly-equivariant model's error is **flat over the entire orbit** (out-of-wedge / in-wedge ratio 1.1–1.2) — a certificate that holds from partial data by the identity $\text{err}(R_\theta s) = \text{err}(s)$. A non-equivariant MLP **scaled** 84× **in parameters** (4k → 337k) buys real *interpolation* — its in-wedge error drops 30–166× and even **beats** the 56×-smaller equivariant model in-distribution — but its error still climbs 170–2700× out-of-wedge, and the largest MLP remains 10–155× worse than the equivariant model there. Scaling 84× barely moves the out-of-wedge floor; the walls never flatten. The fair nuance — *scale can win in-distribution* — is exactly why the contribution is the **certificate** (an orbit-wide guarantee from partial data), not a per-point accuracy contest.

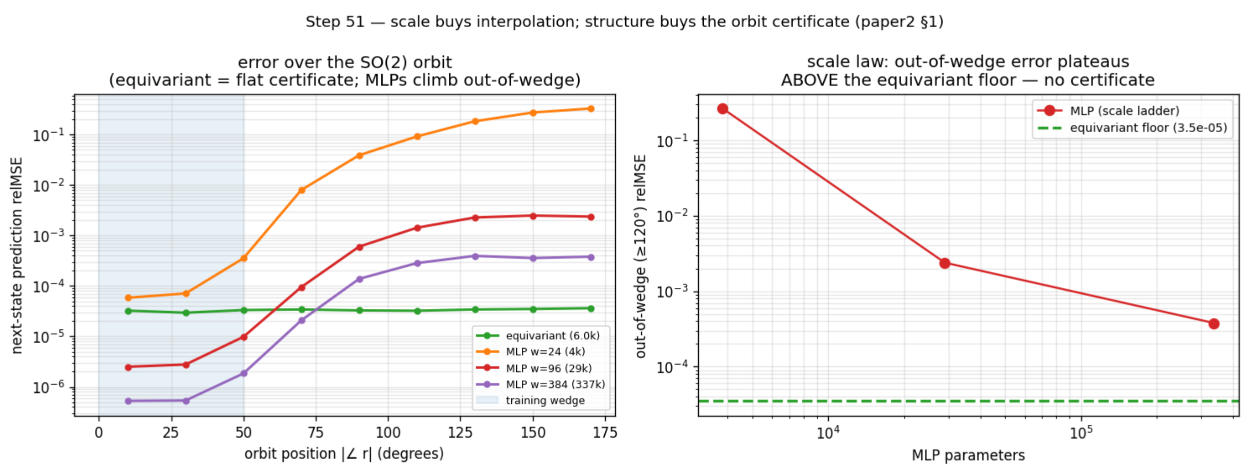

Figure 4: Step 51 — left: error over the SO(2) orbit (equivariant flat; MLPs dip below it in-wedge then climb out); right: out-of-wedge error vs MLP scale plateaus far above the equivariant floor.

4.5 Approximate symmetry — graceful degradation and a measured threshold

Step 53 (3 seeds). We break the *world's* SO(2) symmetry with an anisotropy knob β (the potential becomes $V = \frac{1}{2}(x^2 + (1+2\beta)y^2)$, so angular momentum is no longer conserved) and re-run the wedge-train / full-circle-test. Three findings, robust across seeds: (i) at β=0 the certificate is exact — the equivariant model is 68–320× **better out-of-wedge** than the MLP; (ii) as the measured world symmetry-defect ϵ_{world} grows, the equivariant out-of-wedge error grows **smoothly and monotonically** (correlation +0.88–0.98 — Theorem B's $\epsilon_{\max}$ term, a *graceful* slope, not a cliff); and (iii) the equivariant model keeps beating the non-equivariant one up to a **measured symmetry-content threshold** ($\epsilon_{\text{world}} \approx$ 0.01–0.06), beyond which the (now-wrong) symmetry assumption hurts more than it helps. So *approximate* symmetry buys an *approximate* certificate with exactly the error budget the theory predicts — and the boundary where structure stops paying is itself measured, not assumed.

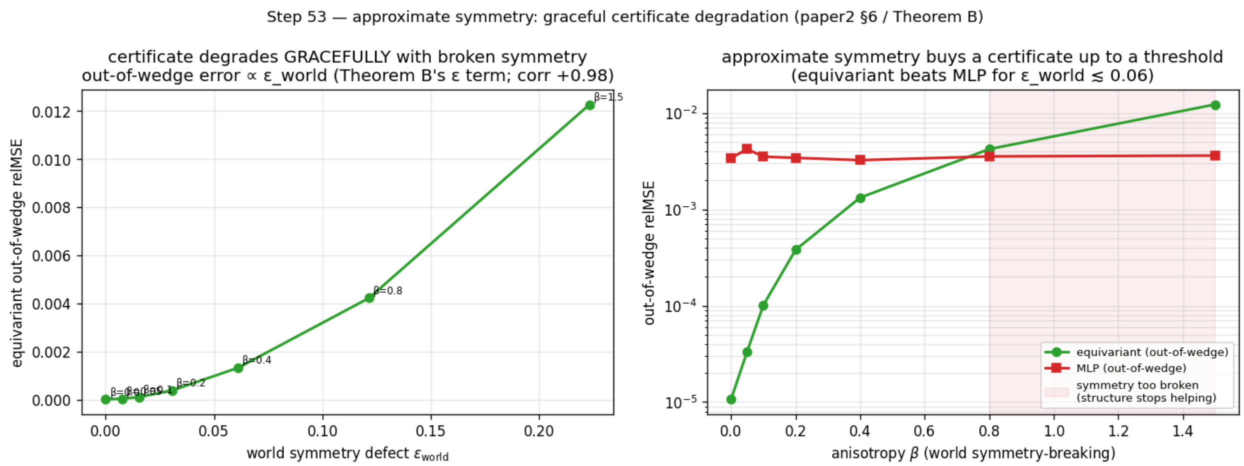

Figure 5: Step 53 — left: equivariant out-of-wedge error grows $\propto \epsilon_{\mathrm{world}}$ (Theorem B's ϵ term); right: the equivariant model beats the MLP out-of-wedge up to a symmetry-content threshold, then crosses over.

---

5. Why this is new (precise delta vs prior work)

The $[A]/[B]/[C]$ device (our prior work). A companion line establishes, for *one* group element and a single resolution, that exact equivariance makes one-step error constant across the group $[B]$, with an exact-flatness $[A]$ and a closed-loop invariance $[C]$. **This paper is the lift of that corner along three axes:** Theorem A is the multi-step, full-monoid generalization of $[B]$; Theorem B adds the horizon×resolution stratification; and the composition-closure lemma turns k generator checks into an exponential certified set. The $[A]/[B]/[C]$ results are the $|w|=1$, single-resolution corner of the certified region.

Equivariant predictors — BRo-JEPA, UWM-JEPA (a corroboration, and a shared caveat). Block-rotation predictors (BRo-JEPA, $\mathbb{Z}/10\mathbb{Z}$ on MNIST, 99.46% zero-shot) and unitary-predictor world models (UWM-JEPA, $U(d)$) independently report striking zero-shot transfer by *matching the predictor to a group structure* — exactly our thesis on different groups. **What they offer us:** cross-group evidence; and our Theorem A explains *why* their zero-shot works (the representation cancels inside the norm — orthogonal for BRo, unitary for UWM) while the closure lemma explains *how far* (the whole generated monoid from the generators — BRo verifies the cyclic case, our Step 49 the exponential $\mathbb{Z}_2^k$ case). **The challenge they raise (which we share and state):** BRo's circular structure is hand-constructed and its predictor shape matches the answer — the same "constructed-teacher" sweetness as our toy worlds (§6). We do not hide behind their numbers; we cite them as a *mechanism* corroboration, not a scale result.

Latent-geometry regularizers — LeJEPA, and UR-JEPA's challenge. LeJEPA pushes the latent toward an isotropic Gaussian (SIGReg); **UR-JEPA (Le, 2026) directly challenges that target**, arguing isotropy conflicts with the manifold hypothesis and that a *data-discovered* anisotropy (uniform rectifiability) is better. This is the sharpest question posed to our latent-geometry line, and we answer it head-on (companion paper, and Step 54/56 here): our latent's anisotropy is neither isotropic (LeJEPA) nor data-discovered (UR-JEPA) but **group-prescribed** — the covariance is *pinned to the representation* ρ ($\rho\Sigma\rho^\top = \Sigma$; measured residual 3×10^{-4} vs a non-equivariant MLP's 1.04). Crucially, we **concede UR-JEPA its point and sharpen ours**: in a head-to-head (Step 56) the data-fit anisotropy *is* best **in-distribution**, but **only the group-prescribed anisotropy transfers out of distribution** ($\sim$320×). The certificate is therefore complementary to both — a *first-order, per-situation* guarantee from the group, not a *second-order* distributional prior — and it tells you precisely *when* a discovered anisotropy will fail to generalize (off the training orbit).

Oracle-bypass diagnostics — IMWM (an independent validation, and a complementary bot-

tleneck). IMWM (Gao et al., 2026) uses the *same* falsifiable move as our Steps 43/47 — **replace the learned model with the exact/oracle dynamics and see what is still missing** — and finds a finite-budget planner *still fails*. **What it offers us:** independent corroboration that "swap in the perfect model" is the right instrument. **The distinction (and our response):** IMWM attributes the residual to **search** (proposal-sampling volume); our oracle-bypass attributes it to **representation** (permutation-invariant encoder pooling). These are *complementary* bottlenecks, not rivals — a predictability certificate bounds the *model's* error over $\langle S\rangle \times T \times \epsilon$; it makes no claim about the planner's search, and IMWM's result is a useful reminder that a perfect certificate still needs an adequate search budget downstream. We state this scope explicitly (§6).

Geometry on the action side — LDA (an ally on geometry, an opponent on generation). LDA ("The Lie We Tell", Chuang et al., 2026) names the **Euclidean Fallacy** — flattening an SE(3) pose into $\mathbb{R}^{12}$ breaks the manifold constraint, the coordinate-change equivariance, and geodesic optimality — and fixes it by running diffusion *on* SE(3) (tangent-space score, exp-map retract). **What it offers us:** a strong external statement of our core motivation — *don't flatten geometric quantities* — now landed on the action/policy side. **The tension it raises:** LDA is a *diffusion policy* (the generative direction this project is contrarian to), so it is simultaneously an ally (geometry) and an opponent (generation); and its gains are "nice, not decisive" (CALVIN task length 3.27 → 3.51, +7.3%), the same *equivariant-method picture* as our Vector-Neuron ×1.36 — modest absolute accuracy, stability under OOD/constraints. Our certificate framing *explains* that picture: a geometric prior buys a *kind* of guarantee (consistency across the group / OOD), not a uniform accuracy jump, so "nice not decisive" average numbers are exactly what one should expect — and the certificate, not the benchmark delta, is the right thing to report.

Predictability horizons (chaos / NWP). The $T(\epsilon) \sim \log(1/\epsilon)/\lambda$ law is classical for dynamical systems; our contribution is to (i) *measure it on a learned latent world model*, (ii) tie its slow subspace to the group-invariant subspace (the Noether hinge), and (iii) fold it into a single multi-axis certificate for equivariant models.

---

6. Limitations and honesty

- **Toy scale.** All five experiments are CPU/1-GPU-scale proofs of principle. We claim a *mechanism and a tool*, not a scaled-SOTA benchmark. The certificate's value is the *kind* of statement it makes, demonstrated cleanly.
- **Approximate symmetry — measured, with the boundary made explicit.** Real-world symmetry is approximate; §4.5 (Step 53) measures the resulting degradation directly — it is **graceful** (out-of-wedge error $\propto \epsilon_{\text{world}}$, exactly Theorem B's $\epsilon_{\max}$ term) and the equivariant certificate remains valuable only **up to a symmetry-content threshold** ($\epsilon_{\text{world}} \approx 0.01$–$0.06$), beyond which structure stops paying. The Noether hinge's **lift to a 3D contact interaction is now partly measured** (§4.3, Step 57): the Noether-content half lifts (R^2=0.86), but the clean containment is 2D-specific because in 3D angular momentum is a conserved ℓ=1 vector — so the open item is now precise: a *3D-aware* containment statement slow $\subseteq$ (invariant $\oplus$ conserved-equivariant). What remains genuinely **not yet done** is lifting the hinge (§3) to the controlled SO(2) system to a contact-rich, approximately-symmetric *embodied* model — the primary next experiment, flagged openly, not hidden.
- **The hinge is a measured conjecture.** Confirmed on a controlled system, with the honest non-converse (invariant $\not\Rightarrow$ slow). We do not claim a proof that learning *must* place conserved quantities in the invariant block — only that, where the dynamics are symmetric, it *does* in our experiments.
- **"Flat" is not "good."** A constant error across the group says nothing about the error's *magnitude*; Theorem A certifies *consistency*, and we report absolute errors alongside ratios throughout.

---

7. Conclusion

For equivariant world models, structure delivers a *kind* of result scaling cannot: a **predictability certificate** — a provable, computable, training-free region across configuration, horizon, and resolution. We proved the master theorem, exhibited the certified region as the coarse-invariant-slow-low-composition corner, measured the Noether hinge that unifies the symmetry and time axes, and showed that an 84×-scaled non-equivariant model buys interpolation but never the certificate. *Scale buys interpolation; structure buys a certificate.* The same criterion that tells you which compositions a robot policy will handle zero-shot also tells you why eclipses are forecastable for millennia and weather is not — one structural law, read across domains.

Reproducibility

Result	Experiment	Test	Seeds	Headline number
Exact-flatness + spectrum	`experiments/step47_certificate.py`	`tests/test_step47_certificate.py`	3	relMSE ×1.00 flat (m=0..8); 48/96 contractive
$\mathbb{Z}_2^6$ exponential cert	`experiments/step49_iching_certificate.py`	—	1	6 generators → all 64; worst relMSE $\sim 10^{-33}$
Noether hinge	`experiments/step50_noether_hinge.py`	`tests/test_step50_noether_hinge.py`	3	R^2 0.92–0.99 vs $\leq$ 0.01; cert 10^{-16} vs 1.17
Structure vs scale	`experiments/step51_structure_vs_scale.py`	—	3	flat 1.1–1.2 vs MLP 170–2700×; 10–155×
Horizon × resolution	`experiments/step52_horizon_resolution.py`	`tests/test_step52_horizon_resolution.py`	3	$\hat{\lambda}$=0.69 vs $\ln 2$; slope $\approx 1/\lambda$
Approximate symmetry	`experiments/step53_approximate_symmetry.py`	—	3	cert exact at β=0 (68–320×); graceful $\propto \epsilon$ (corr 0.88–0.98); threshold $\epsilon \approx$ 0.01–0.06
Embodied/contact hinge lift	`experiments/step57_embodied_hinge.py`	—	1	Noether content lifts (invariant R^2=0.86 vs 0.05); clean containment 2D-specific (3D L is a conserved ℓ=1 vector)

Every experiment sets random seeds explicitly, prints an `INCONCLUSIVE` verdict rather than loosen a gate, and writes its figure + JSON to `papers/figures/`. The full test suite (81 tests) passes together; `tests/conftest.py` isolates the float64 experiments from the float32 codebase.